Zero-Division Containers: Draft of an Arithmetic System

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Status: Draft, work in progress. This is not a complete or verified theory.

Abstract

This draft proposes an arithmetic system in which dividing an integer by zero is a defined operation. The result is not an ordinary number but a new object, the *container* `n|0`, which stores the value `n` together with a zero divisor. The draft presents the objects, the provisional rules, worked examples, and the decisions that are still open.

Motivation

The motivation is to give meaning to dividing an integer by zero without the integer disappearing: the result keeps the original value inside a new object.

Objects

- **Ordinary integers:** `n`, with their usual arithmetic.
- **Containers:** `n|0`, made of an integer value `n` and a zero divisor. Each container has its own identity: they are written `C■`, `C■`, etc. when they need to be told apart.

Provisional rules

1. **Creation.** `n/0` creates a new container with value `n`.
2. **Identity.** Every time a container is created, it is distinct from all previous ones even if it has the same value: `C■ ≠ C■` even if both hold 13.
3. **Preserving addition (⊕).** `n|0 ⊕ m = (n+m)|0`. The container remains.
4. **Extracting addition (+).** `(n|0) + m = n + m`. The container disappears and an integer remains.
5. **Extraction.** `E(n|0) = n`.
6. **Multiplication by zero.** `n × 0 = n|0`. Multiplying by zero does not make `n` disappear: it stores it in a container.
7. **Reappearance.** Dividing by zero an integer obtained by extraction creates a new container (rules 1 and 2).

Worked examples

Example 1. `10|0 ⊕ 3 = 13|0`. The container is preserved.

Example 2. `(13|0) + 2 = 15`. Adding with an integer extracts the value.

Example 3, container cycle. Let `C■ = 13|0`. Then `E(C■) = 13`. Next, `13/0 = C■ = 13|0`, with `C■ ≠ C■`. As long as no integer is added with `+`, the container stays.

Example 4. `10 × 0 = 10|0`. Multiplying by zero keeps the 10 inside a container, just like `10/0 = 10|0`.

Properties to prove

- `E(x ⊕ y) = E(x) + E(y)`.
- `E(n/0) = n`.
- Commutativity and associativity of `⊕`.
- Consistency of evaluation order in mixed expressions.

Open questions

1. **Container identity.** If `C■ ≠ C■` with the same value, we need a formal definition of what tells them apart (for example, a label or a creation counter).
2. **Two equalities.** Equality of value should be separated from equality of structure, with different symbols.
3. **Use of ⊕ and +.** State without ambiguity when each one applies.
4. **Commutativity.** Do `(13|0) + 2` and `2 + (13|0)` give the same result?
5. **Multiplication and division of containers.** What is `(n|0)/0`, and what happens with `(n|0) × m`?
6. **Mixed expressions.** What is `(n|0) × 0`, and in what order are expressions like `12 + 2/0 × 0` evaluated?

Author's note — October 24, 2026

This first draft was written on October 24, 2026, a date that holds personal significance for the author as a Pythagorean milestone. The author marks this day as a meaningful point in her mathematical journey: the beginning of a system she intends to explore, refine, and develop over time.

This document records the ideas and provisional rules established on this date. It is a work in progress, not a verified mathematical theory.

Next steps

1. Formalize the rules with explicit definitions and axioms.
2. Resolve the open questions.
3. Prove or refute the listed properties.

How to cite

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